

Agentic Algebra

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Abstract

We develop a formal theory of *agents* in finite acyclic Bayesian networks. For every subset of variables we consider its Markov blanket and define a closure operator that maps each subset to the maximal interior sharing its exterior. The fixed points of this closure form a Moore family (complete lattice). We prove that the lattice is complemented via the interior–exterior flip, but neither modular nor distributive in general. We introduce a four-way classification of blanket nodes into actuators, sensors, mediators, and priors, and study decomposability conditions under which an agent can be split into independent sub-agents. An information-theoretic measure of agency $M(I) = I(B; I, E)$ quantifies the degree of causal coupling across the Markov blanket and is shown to be subadditive with respect to lattice joins. We discuss the application of this framework to the identification and weighting of moral patients in an empowerment-based AI alignment framework.

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(Scratchpad): Uses of MB in the AIF literature

Ref. [Bru+22]:

- Discussing [Fri13] “Spectral graph theory is then used to identify the eight most densely coupled nodes, which are stipulated to be the “internal” states. Given these internal states, the Markov blanket is then found through tracing the parents, children, and co-parents of children in the network” [...] “proposing that the variables in this Markov blanket can be further separated into “sensory” and “active” states. The sensory states are those states of the Markov blanket whose parents are external states, while the active states are all other states of the Markov blanket (typically, but not always, active states will have children who are external states).”
- “This function is far removed from the initial auxiliary role played by Markov blankets in variational inference, where notions of temporal dynamics and system integrity do not come up. Instead, Markov blankets serve here as a real boundary between organism and world, that is, what we are calling a “Friston blanket.” [...] “Why and how have Markov blankets been reified to act as parts of the target system, for example, by delineating its spatiotemporal boundaries, rather than merely being used as formal tools intended for scientific representation and statistical analysis? When did the map become conflated with the territory?”

Ref [Cla17]:

- “Notice that the mere fact that some creature (a simple feed-forward robot, for example) is not engaging in active online prediction error minimization in no way renders the appeal to a Markov blanket unexplanatory with respect to that creature. The discovery of a Markov blanket indicates the presence of some kind of boundary responsible for those statistical independencies. The crucial thing to notice, however, is that those boundaries are often both malleable (over time) and multiple (at a given time), as we shall see.” (p.4)
- section in the Multi-Agent Risks From Advanced AI report [Ham+25]
- <https://acsresearch.org/>
- Altmann et al. “Emergence in Multi-Agent Systems: A Safety Perspective”

1 Preliminaries and Basic Definitions

1.1 Bayesian Networks and Moral Graphs

Definition 1.1 (Bayesian network). A finite acyclic Bayesian network is a pair (G, P) where $G = (V, E)$ is a directed acyclic graph (DAG) and P is a joint probability distribution over random variables indexed by V that factorises according to G :

$$P(X_V) = \prod_{v \in V} P(X_v \mid X_{\text{pa}(v)}).$$

Definition 1.2 (Moral graph). The *moral graph* of a DAG G is the undirected graph G^m obtained by connecting any two parents of a common child (if not already connected) and then making all edges undirected.

Definition 1.3 (Markov blanket). For a subset $I \subseteq V$, the *Markov blanket* $B(I)$ is the set of nodes not in I that are parents, children, or co-parents (parents of common children) of some node in I . In the moral graph, this is exactly the open neighbourhood:

$$B(I) = \{v \in V \setminus I : \exists u \in I \text{ with } \{u, v\} \text{ an edge in } G^m\}.$$

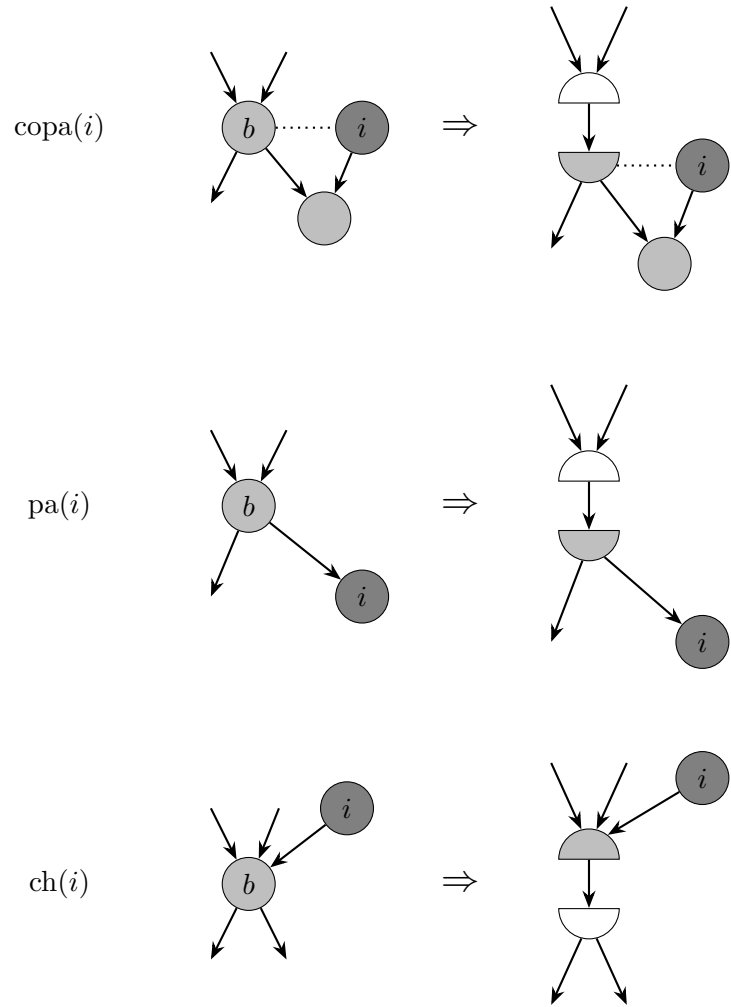


Figure 1: Splitting blanket nodes

Definition 1.4 (Closed neighbourhood and exterior). The *closed neighbourhood* is $N[I] = I \cup B(I)$. The *exterior* is $E(I) = V \setminus N[I]$.

By the global Markov property, $I \perp\!\!\!\perp E(I) \mid B(I)$ for any $I \subseteq V$.

1.2 Motivation: The Markov Blanket Approach to Agent Detection

The Friston school of active inference proposes to identify agents in dynamical systems via their Markov blankets. Given a partition of variables into internal states μ , sensory states s , active states a , and external states x , the blanket $B = s \cup a$ is required to d-separate μ from x . An additional asymmetric condition distinguishes sensory from active states: sensory states have no parents in μ , and active states have no parents in x .

This approach has several well-known difficulties:

- (i) It requires a *candidate* partition as input; the algorithm is a consistency check, not a discovery procedure.
- (ii) The conditional independence structure is symmetric between interior and exterior: $B(I)$ d-separates I from $E(I)$, and equally $B(E(I))$ d-separates $E(I)$ from I . The graph alone cannot determine which side is ‘the agent’. *Jaime: We should adopt a ‘the blanket is part of the agent’ ontology from the start, and (if so), update this section accordingly.*
- (iii) Every subset $I \subseteq V$ yields a valid Markov blanket; the sensory/active decomposition adds no filtering power beyond trivial blanket existence (see ??) *Jaime: but it could help with edge cases, e.g. is an ‘agent’ with no sensory nodes really an agent? Or in the most degenerate case, is a node that has no parents nor children an agent? Do we need to impose that agents have both sensory and active vertices, i.e. $B_{\text{sen}}(I) \neq \emptyset$ and $B_{\text{act}}(I) \neq \emptyset$? Jobst: Good idea for further down!*

We resolve these difficulties by shifting attention from the blanket to the *exterior*: canonically associating to each exterior E the unique maximal interior $I^*(E)$, and studying the lattice of such canonical agents. *Jaime: Focusing on the exterior could be troublesome for large systems (combinatorial explosion). Maybe it is worth aiming to make this framework local by ‘discarding’ any part of the exterior that is not connected to the blanket? (i.e. use a ‘minimal exterior’ comprising a ‘blanket around the blanket’?) See §2.5 below.*

2 The Closure Operator and the Agent Lattice

2.1 Definition of the Closure Operator

Definition 2.1 (Agentic Closure). Let $E : \mathcal{P}(V) \rightarrow \mathcal{P}(V)$ be the *exterior operator* defined by $E(I) = V \setminus N[I]$. We define the *agentic closure* ϕ as the square of the exterior operator:

$$\phi(I) = E(E(I)).$$

A set I is a *canonical interior* (or simply *closed*) if $\phi(I) = I$.

Proposition 2.2 (Closure Properties). *The operator $\phi = E^2$ is a Moore closure operator.*

Proof.

- Monotonicity: Since E is antitone ($I \subseteq J \implies E(J) \subseteq E(I)$), the operator E^2 is monotone.
- Extensivity ($I \subseteq E^2(I)$) follows from the fact that no node in I can be in $E(I)$ or adjacent to $E(I)$.

- Idempotency ($E^2(E^2(I)) = E^2(I)$) follows from the identity $E^3 = E$. To see that $E^3(I) = E(I)$ holds for any $I \subseteq V$, we apply the antitone operator E to the inclusion $I \subseteq E^2(I)$ to obtain $E(I) \supseteq E^3(I)$. Conversely, substituting $E(I)$ into the extensivity rule yields $E(I) \subseteq E^3(I)$. Thus $E^3 = E$, making E^2 a Moore closure operator.

□

2.2 The Lattice of Canonical Interiors

Definition 2.3 (Interior lattice). Let $\mathcal{A} = \{I \subseteq V : \phi(I) = I\}$ denote the set of all closed sets. We call elements of $\mathcal{A}$ *canonical interiors*.

Corollary 2.4. $\mathcal{A}$ is a Moore family (closure system). It is a complete lattice under set inclusion, with meet and join

$$\bigwedge_{\alpha} I_{\alpha} = \bigcap_{\alpha} I_{\alpha}, \quad \bigvee_{\alpha} I_{\alpha} = \phi\left(\bigcup_{\alpha} I_{\alpha}\right).$$

The bottom element is $\emptyset$ and the top element is V .

Proof. The closure of any intersection of closed sets is itself, since ϕ is a closure operator on a finite set. The meet and join formulas are standard for Moore families (see e.g. [DP02]). □

2.3 The Exterior Map

Before formalising the agent as a physical entity, we establish two theorems governing the behaviour of the exterior under lattice operations.

Theorem 2.5 (Exterior is anti-isomorphic to interior). For $I_1, I_2 \in \mathcal{A}$:

$$I_1 \subseteq I_2 \iff E(I_1) \supseteq E(I_2).$$

Proof. ($\Rightarrow$): $I_1 \subseteq I_2$ implies $N[I_1] \subseteq N[I_2]$ implies $E(I_1) \supseteq E(I_2)$.

($\Leftarrow$): Suppose $E_2 \subseteq E_1$ but $\exists i \in I_1^* \setminus I_2^*$. Since $i \notin E_2 \subseteq E_1$ and $i \notin I_2^*$, we have $i \in B_2$. Take any neighbour n of i (parent, child, or co-parent via shared child c). If n were in $E_2 \subseteq E_1$, then since $i \in I_1^*$, node n would be in $B(I_1^*) \cup I_1^*$, contradicting $n \in E_1$. Hence all neighbours of i avoid E_2 , so $N[i] \subseteq V \setminus E_2$, meaning $i \in I^*(E_2) = I_2^*$, a contradiction. □

Theorem 2.6 (Exterior of a join). For any family of closed sets $I_{\alpha} \in \mathcal{A}$,

$$E\left(\bigvee_{\alpha} I_{\alpha}\right) = \bigcap_{\alpha} E(I_{\alpha}).$$

Proof. Let $J = \bigvee_{\alpha} I_{\alpha}$. Since $E(J) = V \setminus N[J]$ and $N[J] = \bigcup_{\alpha} N[I_{\alpha}]$ (the closed neighbourhood of the closure equals the union of the closed neighbourhoods, since no exterior node enters the blanket under union), we obtain $E(J) = V \setminus \bigcup_{\alpha} N[I_{\alpha}] = \bigcap_{\alpha} (V \setminus N[I_{\alpha}]) = \bigcap_{\alpha} E(I_{\alpha})$. □

2.4 The Agentic Isomorphism

While the lattice $\mathcal{A}$ elegantly captures the topology of canonical interiors, an *agent* is constituted by both its internal states and the sensory-active blanket that mediates its relationship with the world. We therefore shift our operational focus from the interior I to the total physical footprint of the agent.

Definition 2.7 (The Canonical Agent). For any canonical interior $I \in \mathcal{A}$, the corresponding *canonical agent* $A(I)$ is defined as its closed neighbourhood:

$$A(I) = N[I] = I \cup B(I).$$

Equivalently, the agent is the strict complement of its exterior: $A(I) = V \setminus E(I)$.

Because the exterior map E is a bijection on the canonical interiors (Theorem 2.5), the mapping $I \mapsto A(I)$ is also a bijection. This allows us to define an isomorphic lattice where the elements are agents rather than interiors.

Definition 2.8 (The Agent Lattice). Let $\mathcal{A}^* = \{A(I) : I \in \mathcal{A}\}$ denote the lattice of canonical agents. The lattice operations are strictly induced by the underlying interiors:

$$\begin{aligned} A_1 \wedge^* A_2 &= A(I_1 \wedge I_2) \\ A_1 \vee^* A_2 &= A(I_1 \vee I_2) \end{aligned}$$

By elevating the lattice to the total footprint A , the operation of macro-agent assembly (the join) resolves into a remarkably simple physical property.

Theorem 2.9 (The Footprint of a Join). *In the lattice of canonical agents $\mathcal{A}^*$, the agentic join is strictly equal to the set-theoretic union:*

$$A_1 \vee^* A_2 = A_1 \cup A_2$$

Proof. Let $I_{12} = I_1 \vee I_2$ be the join of the underlying interiors. By Theorem 2.6, the exterior of the join is the intersection of the individual exteriors: $E(I_{12}) = E(I_1) \cap E(I_2)$. By Theorem 2.7, any canonical agent is the complement of its exterior, $A = V \setminus E(I)$. Therefore, the physical footprint of the joined agent is:

$$A(I_{12}) = V \setminus E(I_{12}) = V \setminus (E(I_1) \cap E(I_2))$$

Applying De Morgan's laws yields:

$$A(I_{12}) = (V \setminus E(I_1)) \cup (V \setminus E(I_2)) = A_1 \cup A_2. \quad \square$$

Remark 2.10 (Internal Reorganisation vs. External Expansion). Theorem 2.9 demonstrates that when two agents merge, they do not indiscriminately absorb arbitrary nodes from the deep environment to form their new boundary. Their combined physical footprint is strictly bounded by the union of their individual footprints. As we will see below, the non-linear complexity of the join arises entirely through *internal reorganisation*—specifically, the internalisation of shared boundary nodes—which occurs strictly within the outer envelope of $A_1 \cup A_2$.

2.5 The Habitat and Internal Reorganisation

The exterior $E(I)$ of a canonical agent can encompass the entire universe separated by the blanket. To formalise the local environment with which an agent directly interacts, we define its habitat.

Definition 2.11 (Habitat). For a canonical agent $A \in \mathcal{A}^*$, the *habitat* (or *halo*) $H(A)$ is the open neighbourhood of the agent within the exterior:

$$H(A) = \{e \in V \setminus A : \exists a \in A \text{ with } \{e, a\} \text{ an edge in } G^m\}.$$

Proposition 2.12 (Localised Habitat of a Join). *For any two agents $A_1, A_2 \in \mathcal{A}^*$, the habitat of their join $A_{12} = A_1 \vee^* A_2$ is determined locally:*

$$H(A_{12}) = (H(A_1) \setminus A_2) \cup (H(A_2) \setminus A_1).$$

Proof. Because $A_{12} = A_1 \cup A_2$, the exterior of the join is $(V \setminus A_1) \cap (V \setminus A_2)$. The neighbours of the union in this exterior are simply the original habitats, minus any nodes that have been enveloped by the footprint of the other agent. $\square$

Theorem 2.13 (Internal Reorganisation and Blanket Absorption). *Let $A_1, A_2 \in \mathcal{A}^*$ be agents with interiors I_1, I_2 and boundaries B_1, B_2 . When joined into $A_{12} = A_1 \cup A_2$, the new interior I_{12} is larger than the union of the original interiors, explicitly absorbing any boundary nodes that lose connection to the joint habitat:*

$$I_{12} = I_1 \cup I_2 \cup \{b \in B_1 \cup B_2 : N(b) \cap H(A_{12}) = \emptyset\}.$$

Consequently, the new boundary is strictly a subset of the original combined boundaries:

$$B(I_{12}) = (B_1 \cup B_2) \setminus I_{12}.$$

Proof. Because $A_{12} = A_1 \cup A_2$, we have $I_{12} \cup B(I_{12}) = I_1 \cup B_1 \cup I_2 \cup B_2$. The nodes $I_1 \cup I_2$ are already structurally separated from the exterior, so they must belong to I_{12} . The remaining nodes are the combined boundaries $B_1 \cup B_2$. A node $b \in B_1 \cup B_2$ remains in the boundary of the joined agent if and only if it maintains an edge to the exterior $V \setminus A_{12}$. By definition of the habitat, any such edge must connect to $H(A_{12})$. If no such connection exists, b is topologically isolated from the exterior and thus satisfies the condition for interiority, joining I_{12} . $\square$

Remark 2.14 (Habitat Smothering). This result provides a structural mechanism for macro-agent assembly. When agents merge, boundary nodes that were previously mediating between the agents (e.g., a sensor in A_1 reading an actuator in A_2) are ‘smothered’. Deprived of a connection to the remaining external environment $H(A_{12})$, they are reclassified by the closure operator as internal computational nodes of the new macroscopic agent.

3 Complement and Non-Modularity

3.1 The Complement and Agent-Environment Duality

In the lattice of canonical interiors $\mathcal{A}$, the complement of an interior is simply its exterior: $I^c = E(I)$. We now lift this operation to the lattice of canonical agents $\mathcal{A}^*$ to examine the physical relationship between an agent and its environment.

Definition 3.1 (Agentic Complement). For an agent $A \in \mathcal{A}^*$ generated by interior I , we define its complement as the agent generated by the complement interior:

$$A^c = A(I^c)$$

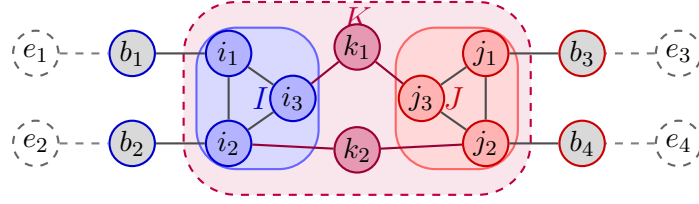
Theorem 3.2 (The Shared Membrane). *For any canonical agent $A \in \mathcal{A}^*$ with interior I and boundary $B(I)$:*

- (i) $A^c = V \setminus I$ (the environment encompasses everything except the agent’s internal states);
- (ii) $A \cup A^c = V$ (the agent and environment partition the universe);
- (iii) $A \cap A^c = B(I)$ (the agent and environment intersect exactly at the blanket).

Proof.

- (i) By definition, $A^c = A(E(I))$. Because $E(I)$ is a valid canonical interior (Theorem 2.5), its agentic footprint is its closed neighbourhood: $A(E(I)) = E(I) \cup B(E(I))$. By the symmetry of d-separation in the moral graph, the boundary of the exterior is identical to the boundary of the interior: $B(E(I)) = B(I)$. Thus, $A^c = E(I) \cup B(I)$. Because V partitions strictly into $I \sqcup B(I) \sqcup E(I)$, we have $E(I) \cup B(I) = V \setminus I$.

(a) **Complete isolation:** k_1 and k_2 are absorbed into K



(b) **External connection:** e_5 prevents k_2 from being absorbed

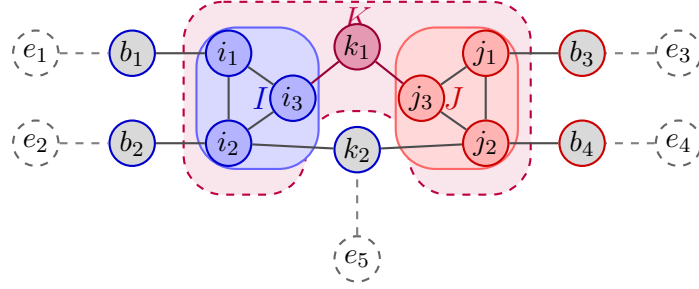


Figure 2: Diagram illustrating Theorem 2.13. The physical footprint of the joined agent is strictly the union $A_1 \cup A_2$. In **(a)**, boundary nodes k_1 and k_2 have no exterior links to the combined habitat, so they are absorbed into the new interior I_{12} . In **(b)**, the addition of e_5 connected to k_2 provides an anchor to the habitat $H(A_{12})$. Thus, k_2 is **not** absorbed and remains part of the macroscopic boundary $B(I_{12})$. **Legend:**

- ●: Nodes forming the initial interiors I_1 and I_2 .
- ○: Nodes in the new boundary $B(I_{12})$ (must have exterior connections).
- : Boundary nodes absorbed into the new interior I_{12} (habitat smothered).
- ⌋: Nodes in the joint habitat $H(A_{12})$.

- (ii) We expand the union: $A \cup A^c = (I \cup B(I)) \cup (E(I) \cup B(I)) = I \cup B(I) \cup E(I) = V$.
- (iii) We expand the intersection: $A \cap A^c = (I \cup B(I)) \cap (E(I) \cup B(I))$. Because I and $E(I)$ are strictly disjoint by definition, the intersection reduces exactly to $B(I)$. $\square$

Remark 3.3 (The Boundary as a Dual Structure). Theorem 3.2 mathematically formalises the cybernetic concept of the Markov blanket. In the A -centric ontology, the blanket $B(I)$ is not merely an external layer surrounding the agent; it is the strict mathematical intersection of the agent and its environment. The boundary is a dual structure, belonging equally to A and A^c , serving as the physical membrane where internal dynamics and external dynamics are structurally coupled.

3.2 Non-Modularity

While the lattice $\mathcal{A}^*$ possesses a complement, it lacks the linear behaviour of a standard topology.

Theorem 3.4. *The lattice of canonical agents $\mathcal{A}^*$ is not modular in general.*

Proof. Consider the path graph $G^m : 1 - 2 - 3 - 4 - 5$. We define three canonical agents based on their interiors:

- Let $I_x = \{5\}$. Its boundary is $B_x = \{4\}$, so $A_x = \{4, 5\}$.
- Let $I_y = \{1, 2\}$. Its boundary is $B_y = \{3\}$, so $A_y = \{1, 2, 3\}$.
- Let $I_z = \{3, 4, 5\}$. Its boundary is $B_z = \{2\}$, so $A_z = \{2, 3, 4, 5\}$.

We have $A_x \leq A_z$ (since $\{4, 5\} \subseteq \{2, 3, 4, 5\}$). For the modular law to hold, we must have $A_x \vee^* (A_y \wedge^* A_z) = (A_x \vee^* A_y) \wedge^* A_z$.

First, compute the left hand side. The meet $A_y \wedge^* A_z$ corresponds to the interior $I_y \cap I_z = \{1, 2\} \cap \{3, 4, 5\} = \emptyset$, which yields the empty agent $\emptyset$. Thus, $A_x \vee^* \emptyset = A_x = \{4, 5\}$.

Next, compute the right hand side. The join $A_x \vee^* A_y$ joins interiors $\{5\}$ and $\{1, 2\}$. Their union $\{1, 2, 5\}$ has a candidate blanket $\{3, 4\}$. However, neither 3 nor 4 has any connections to an exterior node (the habitat is empty). By Theorem 2.13, they are absorbed into the interior. The closure is therefore the entire graph V . Thus, $(A_x \vee^* A_y) \wedge^* A_z = V \wedge^* A_z = A_z = \{2, 3, 4, 5\}$.

Since $\{4, 5\} \neq \{2, 3, 4, 5\}$, the modular law fails. $\square$

Corollary 3.5. *The lattice $\mathcal{A}^*$ is not distributive in general.*

Remark 3.6 (Habitat Smothering and Non-Modularity). The failure of modularity in Theorem 3.4 is a direct consequence of habitat smothering. In the path graph example, the agents A_x and A_y are spatially separated by nodes 3 and 4. While node 3 acts as the habitat for A_x , and node 4 acts as the habitat for A_y , their combination traps the intervening space. The structural tension is lost; nodes 3 and 4 are internalised, and the boundary collapses, blowing up the combined agent's physical footprint to encompass the entire graph. Non-modularity is thus the mathematical signature of the non-linear topological phase transitions that occur when distant agents merge and consume their shared environment.

4 Classification of the Agentic Interface

Summary of the process for a candidate set The sequence to formalise a canonical agent A^* from an arbitrary seed interior $I \subseteq V$ is as follows:

1. **Initial Shelling (Blanket):** Determine the candidate blanket $B(I)$ by finding the open neighbourhood of I in the moral graph G^m . This establishes the candidate physical footprint $A = I \cup B(I)$.

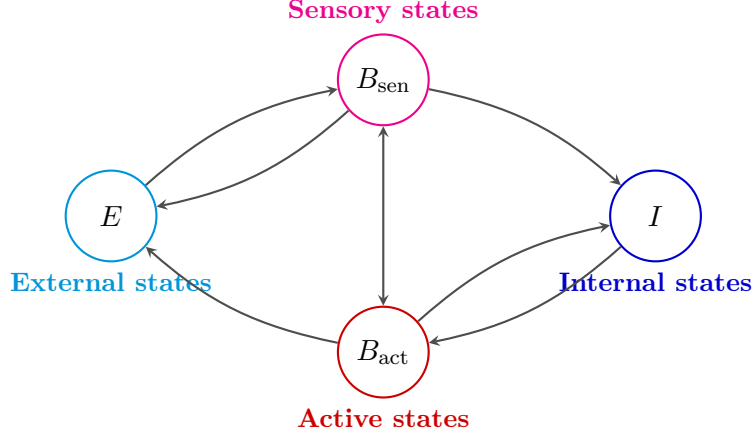


Figure 3: Perception-action loop in active inference. [Note: In physical formulations of active inference, the arrow from sensory to external states ($B_{\text{sen}} \rightarrow E$) represents the necessary physical back-reaction of sensing, as biological sensors inevitably interact with and alter the environment they measure. Mathematically, this reflects the fact that the continuous dynamics of external states are coupled to the entire Markov blanket, encompassing both sensory and active states.]

2. **Secondary Shelling (Habitat):** Determine the **habitat** $H(A)$ by finding the neighbours of the agent that lie in the exterior $E = V \setminus A$. This represents the only part of the universe that the candidate agent can ‘sense’ or ‘affect’ directly.
3. **Agentic Closure (Internal Reorganisation):** Compute the canonical interior I^* by internalising any boundary nodes isolated from the habitat. A node $b \in B(I)$ is absorbed into I^* if and only if it has no structural links to $H(A)$:

$$I^* = I \cup \{b \in B(I) : N(b) \cap H(A) = \emptyset\}.$$

This procedure yields the canonical agent $A^* = I^* \cup B(I^*)$.

4. **Habitat Persistence:** Because the closure operator internalises nodes without expanding the physical footprint A , the exterior remains identical. Consequently, the habitat remains stable: $H(A^*) = H(A)$. The closure operator simply ‘tucks away’ internal plumbing without altering the agent’s environmental interface.
5. **Canonical Membrane Determination:** The canonical boundary $B(I^*)$ is the set of non-absorbed nodes from the original candidate; thus, $B(I^*) \subseteq B(I)$. By construction, every node $b \in B(I^*)$ is a topological bridge, possessing (in the moral graph) at least one neighbour in the interior I^* and at least one neighbour in the habitat $H(A^*)$.
6. **Directed Functional Validation:** While every boundary node is a structural bridge in the moral graph, its causal role depends on the underlying DAG. We use the parent/child sets relative to the exterior E to classify these bridges. Jaime: [TODO: do we make all canonical agents valid, or do we filter some out (e.g. disconnected from the rest of V , or if they have no actuators, etc.)?]

4.1 An Environment-Centric Taxonomy

To categorise boundary nodes exhaustively while guaranteeing mathematical monotonicity under lattice operations, we treat the canonical blanket $B(I^*)$ as a physical membrane defined strictly by its relationship to the environment.

Rather than classifying nodes by their internal causal history (which shifts dynamically under joins), we classify them by their directed topological links to the exterior E . For any boundary node $b \in B(I^*)$, we define two environmental Boolean indicators:

$$P_E \equiv \text{pa}(b) \cap E \neq \emptyset, \quad C_E \equiv \text{ch}(b) \cap E \neq \emptyset$$

This yields a symmetric, four-way partition of the agent's causal surface (summarised in Table 1).

1. **Pure Actuators** (B_{act}): $\neg P_E \wedge C_E$
Nodes that drive the exterior without receiving direct environmental input. From the perspective of the habitat, these are the origins of the agent's physical footprint. (This encompasses both deliberate actions driven by the interior, and exogenous boundary clocks).
2. **Pure Sensors** (B_{sen}): $P_E \wedge \neg C_E$
Nodes driven by the exterior that do not project back out. These are the sensory receptors and environmental sinks where external causality terminates on the agent's membrane.
3. **Transceivers / Reflexes** (B_{trans}): $P_E \wedge C_E$
Nodes that are driven by the world and structurally project a physical reflection back into the environment.
4. **Insulators / Structural Mediators** (B_{ins}): $\neg P_E \wedge \neg C_E$
Nodes lacking any directed signals to or from E . These survive the closure operator purely via undirected co-parent edges (v-structures) with the exterior. They act as structural scaffolding where internal logic and external inertia meet, creating structural tension without directed communication.

Input \ Output	$\neg C_E$ (No external projection)	C_E (Projects to exterior)
$\neg P_E$ (No external input)	Insulator / Mediator (B_{ins})	Pure Actuator (B_{act})
P_E (Driven by exterior)	Pure Sensor (B_{sen})	Transceiver / Reflex (B_{trans})

Table 1: The environmental interface taxonomy. By classifying boundary nodes strictly by their directed connections to the exterior $E(A)$, we ensure the classification is monotonically non-increasing under lattice joins. Pure internal loops ($\neg C_E \wedge \neg P_E$ lacking co-parent anchors) and pure environmental loops (lacking I anchors) are topologically swept into the interior I^* or exterior E , respectively, leaving only this true causal interface.

Remark 4.1 (The Pruning of Internal Dynamics and Proprioception). By explicitly dropping internal indicators (P_I, C_I) from the structural taxonomy, we naturally resolve the ambiguity of proprioception. A node driven by the agent (P_I) that reads its own state (C_I) without touching the environment ($\neg C_E \wedge \neg P_E$) is topologically absorbed into the canonical interior I^* by the closure operator. It is an internal computation, not a boundary condition. If a node is an Actuator (C_E) and also projects inward (C_I), it simply possesses the property of proprioception, but its structural role as an Actuator on the agent's physical membrane remains pristine.

5 Decomposability and Sub-Agents

5.1 Separability

The simplest and most robust form of decomposability arises when a macroscopic agent is physically fractured in the moral graph. Because the canonical agent $A = I \cup B(I)$ encompasses

the entirety of the system’s internal and boundary states, we can evaluate its decomposability through strict graph connectivity.

Definition 5.1 (Separability). Two canonical agents $A_1, A_2 \in \mathcal{A}^*$ are *separable* if their physical footprints are strictly disjoint ($A_1 \cap A_2 = \emptyset$) and there are no edges between them in the moral graph G^m .

Theorem 5.2 (Disconnected Agents). *Let $A \in \mathcal{A}^*$ be a canonical agent. If the subgraph induced by A in G^m is disconnected, then A is the join of separable sub-agents. Specifically, if $A_1, \dots, A_k$ are the connected components of A , then each A_i is a valid canonical agent, and:*

$$A = \bigvee_{i=1}^k A_i = \bigcup_{i=1}^k A_i$$

Furthermore, because the components are separable, their internal and boundary nodes partition cleanly: $I = \bigsqcup I_i$ and $B(I) = \bigsqcup B(I_i)$.

Proof. Because A is closed, no boundary node in A can be absorbed by the exterior. Since the components A_i share no edges with one another, the presence or absence of A_j has no topological effect on the exterior connections of A_i . Thus, the local habitat of each component remains strictly independent, meaning no habitat smothering occurs. The agentic closure of each component is simply itself, and their join evaluates strictly to their disjoint union. $\square$

Remark 5.3 (Functional Stability). When an agent is separable, there is ‘something in between’ the sub-agents—specifically, physical distance mediated by the exterior environment E . Because there are no direct edges between A_1 and A_2 , separating them does not sever any direct causal links. Consequently, the functional classification of their boundaries (Table 1) remains perfectly stable. No node gains or loses a parent or child in the transition from the macroscopic agent to the separated sub-agents.

5.2 Weak Decomposability and Shared Interfaces

When an agent is connected in the moral graph, it cannot be separated into non-interacting components. However, it may still be decomposed into distinct computational cores that physically interact.

Definition 5.4 (Weak Decomposability). A connected agent $A \in \mathcal{A}^*$ is *weakly decomposable* if it can be expressed as the join of two valid canonical agents ($A = A_1 \vee^* A_2$) whose interiors are strictly disjoint ($I_1 \cap I_2 = \emptyset$), but whose boundaries overlap:

$$A_1 \cap A_2 = B(I_1) \cap B(I_2) \neq \emptyset.$$

In a weakly decomposable system, the agents are computationally distinct (lacking shared internal states) but physically touch, sharing a sensory or actuating membrane. Because they share boundary nodes, joining them can trigger habitat smothering (Theorem 2.13), meaning the macroscopic interior I_{12} may be strictly larger than $I_1 \cup I_2$.

5.3 The Sub-Agent Relations

We can formalise the relationship between a component and the macroscopic whole using nested inclusion.

Definition 5.5 (Strict Subagent). For agents $A_{\text{sub}}, A_{\text{macro}} \in \mathcal{A}^*$, the agent A_{sub} is a *strict subagent* of A_{macro} , written $A_{\text{sub}} \prec_s A_{\text{macro}}$, if the subagent is entirely contained within the interior of the macro-agent:

$$A_{\text{sub}} \subseteq I_{\text{macro}}.$$

Proposition 5.6 (Environmental Encapsulation). *If $A_{\text{sub}} \prec_s A_{\text{macro}}$, the subagent is cybernetically isolated from the macro-environment. Its habitat is entirely contained within the macro-agent’s internal states:*

$$H(A_{\text{sub}}) \subseteq I_{\text{macro}}.$$

Proof. By definition, $H(A_{\text{sub}})$ consists of the nodes adjacent to A_{sub} that lie outside A_{sub} . Because $A_{\text{sub}} \subseteq I_{\text{macro}}$, any node adjacent to A_{sub} must belong to $I_{\text{macro}} \cup B(I_{\text{macro}})$. However, a node in $B(I_{\text{macro}})$ cannot be adjacent to A_{sub} , because the closed neighbourhood of A_{sub} must remain within I_{macro} by the definition of the canonical interior. Therefore, all neighbours of A_{sub} reside strictly in I_{macro} . $\square$

Observation 5.7 (The ‘Internal Organ’ Analogy). A strict subagent acts as an internal organ. Because its habitat never touches the true exterior $E(A_{\text{macro}})$, its sensors and actuators interact exclusively with the ‘guts’ of the macro-agent. It represents a nested software process or biological subunit lacking direct environmental I/O.

Definition 5.8 (Weak Subagent). For $A_{\text{sub}}, A_{\text{macro}} \in \mathcal{A}^*$, we say A_{sub} is a *weak subagent* of A_{macro} , written $A_{\text{sub}} \prec_w A_{\text{macro}}$, if $A_{\text{sub}} \subsetneq A_{\text{macro}}$ and they share a physical boundary:

$$A_{\text{sub}} \cap B(I_{\text{macro}}) \neq \emptyset.$$

Remark 5.9 (Functional Shift and ‘The Rest of the Agent’). Unlike separable agents, weak subagents experience functional instability. When evaluating the boundary of A_{sub} , its ‘exterior’ includes not only the true environment $E(A_{\text{macro}})$, but also the massive internal buffer of the macro-agent, $A_{\text{macro}} \setminus A_{\text{sub}}$.

If b is a Pure Actuator for the macro-agent ($b \in B_{\text{act}}$), it projects into the environment but its driving parents reside in I_{macro} . From the perspective of the subagent, these parent nodes belong to the exterior. Thus, the node gains external parents (P_E) relative to the subagent, shifting its function to a Transceiver (B_{trans}). Consequently, a weak subagent may lack deliberate control over specific channels of the macro-agent’s interface, as it is buffeted by ‘noise’ generated by the rest of the macro-agent’s interior.

Jaime: Could a ‘distributed agent’ $A_{\text{dist}} \in \mathcal{A}^*$ exist, where a group of agents (agent interiors, actually) $\{A_j\} \subset \mathcal{A}$ is an *open cover* of A_{dist} ?

6 Dynamic Extensions and Spacetime Agents

The framework developed thus far operates on static spatial graphs. However, true agency is intrinsically diachronic: it is characterised by the persistent control of boundaries over time. Applying the closure operator to an instantaneous cross-section of a dynamical system conflates synchronous correlations with causal control. To resolve this, we map our spatial algebra onto a spacetime topology.

6.1 The Unrolled Spacetime Graph

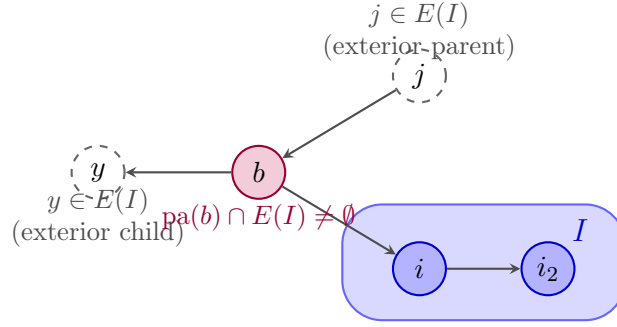
We model the dynamical system as a finite, discrete-time Dynamic Bayesian Network (DBN) unrolled over a temporal horizon $t \in \{0, \dots, T\}$.

Definition 6.1 (Unrolled DBN). Let V be the set of system variables. The unrolled graph consists of nodes $V_{0:T} = \bigcup_{t=0}^T V_t$. The directed edges are partitioned into two sets:

- (i) **Synchronous edges:** Directed links within a single time slice, $V_t \times V_t$.
- (ii) **Diachronic edges:** Directed links spanning consecutive time slices, representing memory or physical persistence, $V_t \times V_{t+1}$.

By definition, the unrolled DBN is a valid DAG, which can be moralised into $G_{0:T}^m$.

(a) Perspective of Subagent I : $b \in B_{\text{trans}}(I)$



(b) Perspective of Macro-agent J : $b \in B_{\text{act}}(J)$

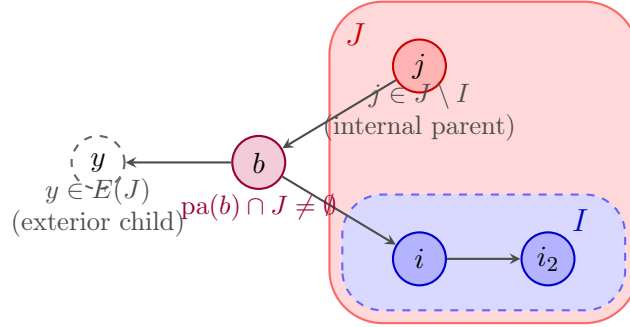


Figure 4: Functional shift across the weak-subagent relation (illustrating Remark 5.9). Node b serves as a shared boundary interface, but its cybernetic classification inverts depending on the reference agent. **(a)** From the perspective of the subagent A_{sub} , the parent node j lies in the local exterior. Because b is driven by this local exterior and projects to the true exterior (y), it functions as a Transceiver/Reflex (B_{trans}) for the subagent. **(b)** From the perspective of the macro-agent A_{macro} , node j is internalised. Because b is now driven purely by the interior and projects to the true exterior (y), it functions as a Pure Actuator (B_{act}).

6.2 The Spacetime Worm

Because the unrolled graph is topologically identical to a static spatial graph, the lattice of canonical agents $\mathcal{A}^*$ applies natively to sequences of states over time.

Definition 6.2 (The Spacetime Agent). An agent $A_{1:T-1}$ is defined as a sequence of physical footprints across time. To avoid the degenerate boundary conditions of initial priors ($t = 0$) and terminal sinks ($t = T$), we restrict the candidate footprint to $t \in \{1, \dots, T-1\}$. We apply the agentic closure operator directly to this temporal sequence. If $A_{1:T-1}$ is a closed canonical agent in $\mathcal{A}^*$, the sequence constitutes a *canonical spacetime worm*.

6.3 Temporal Membrane Partition

When the moral graph $G_{0:T}^m$ is constructed, diachronic edges introduce temporal cross-coupling. If an internal state i_{t+1} is caused by both its past state i_t and a concurrent environmental state e_t , the moralisation connects i_t to e_t via an undirected co-parent edge. Consequently, the membrane $B(I_{1:T-1})$ expands across the temporal axis.

Theorem 6.3 (Spacetime Membrane Partition). *For a canonical spacetime agent $A = A_{1:T-1}$, its physical membrane $B(I)$ partitions strictly into three structural regions:*

$$B(I) = B_{\text{past}} \sqcup B_{\text{spatial}} \sqcup B_{\text{future}} \quad (1)$$

where:

- **Initial Conditions** (B_{past}): Nodes at $t = 0$ that project directed edges into the footprint at $t = 1$. These act as temporal sources or absolute environmental priors.
- **Terminal Sinks** (B_{future}): Nodes at $t = T$ that receive directed edges from the footprint at $t = T-1$. These represent the permanent environmental legacy of the agent.
- **Spatial Boundary** (B_{spatial}): Nodes residing in $t \in \{1, \dots, T-1\}$ that mediate ongoing interactions with the concurrent habitat $H_{1:T-1}$. This set maps directly to our 4-class functional taxonomy (Pure Sensors, Pure Actuators, Transceivers, Insulators) formalised in Table 1.

Proof. The partition follows directly from the restriction of the footprint to $t \in \{1, \dots, T-1\}$. Boundary nodes strictly prior to the interior ($t = 0$) cannot possess children within the agent without violating the temporal arrow, restricting them to source nodes. Boundary nodes strictly posterior ($t = T$) cannot possess parents within the agent, restricting them to sink nodes. All other boundary nodes share time slices with the agent and are subject to the lateral spatial taxonomy. $\square$

6.4 Temporal Separability and Agent Persistence

By applying the decomposability theorems from Section 5 to the temporal axis, we can formally define the lifespan of an agent. An agent persists over time if its consecutive temporal slices cannot be separated.

Definition 6.4 (Temporal Continuation). Let A_t be a canonical agent slice at time t . We say A_t *continues* to $t+1$ if there exists a canonical agent A_{t+1} such that their diachronic join $A_t \vee^* A_{t+1}$ is strongly connected over time (i.e., not separable).

Remark 6.5 (The Death of an Agent). Agentic ‘death’ or dissolution occurs when A_{t+1} becomes separable from A_t . This manifests structurally when the nodes comprising the agent at t lose their causal grip on the graph at $t+1$. Deprived of diachronic edges bridging the consecutive time steps, the agentic closure operator will either absorb the disjoint future nodes into the macroscopic environment E_{t+1} or shatter them into isolated noise, formalising the physical breakdown of the agent’s persistence.

6.5 Algorithmic Tractability and Markovian Extrusion

For large networks or long temporal horizons T , computing the global agentic closure over the entire unrolled DBN becomes computationally intractable. Furthermore, physical agents do not compute their persistence by referencing their absolute future; they persist sequentially.

To bridge the gap between global theoretical limits and computational tractability, we approximate the spacetime worm using a step-by-step algorithm termed *Markovian Extrusion*.

Definition 6.6 (Markovian Extrusion). Let A_t^* be a canonical agent strictly evaluated on the synchronous slice V_t , such that $\phi_{\text{synch}}(I_t^*) = I_t^*$. The Markovian extrusion to $t + 1$ proceeds in two steps:

- (1) **Causal Projection:** Define a candidate footprint A'_{t+1} by collecting all nodes at $t + 1$ that are driven by the agent at t , representing the structural memory of the agent:

$$A'_{t+1} = \{v \in V_{t+1} : \text{pa}(v) \cap A_t^* \neq \emptyset\}$$

- (2) **Synchronic Re-closure:** Apply the spatial agentic closure to the projected candidate to re-establish the boundary against the new environmental state:

$$I_{t+1}^* = \phi_{\text{synch}}(A'_{t+1}) \setminus B_{\text{synch}}$$

Yielding the new footprint A_{t+1}^* .

Remark 6.7 (Global vs. Greedy Closure). The sequence $\{A_1^*, A_2^*, \dots, A_{T-1}^*\}$ generated by Markovian extrusion represents a greedy approximation of the global spacetime worm. Because the synchronic closure lacks access to future habitat states ($H_{>t}$), the extruded agent cannot perform backward absorption (e.g., internalising a node at t because its causal descendants lose habitat connectivity at $t + 2$). Consequently, the extruded physical footprint may be slightly larger than the optimal global footprint. However, it provides a computationally tractable, causal-time algorithm for tracking agentic boundaries in dynamic simulations.

7 Information-Theoretic Agency

With the physical footprint of the agent extended into a spacetime worm (Section 6), we can move beyond static structural correlations. True agency is not merely the synchronous correlation between an interior and a boundary; it is the active, diachronic control of that boundary against the inertia of the environment.

We assume the unrolled nodes $V_{0:T}$ are discrete random variables with a known joint distribution $P(V_{0:T})$.

7.1 Diachronic Agency and Directed Information

To measure dynamic control, we isolate the information flowing strictly forward in time from the agent’s internal logic to its future physical membrane.

Definition 7.1 (Diachronic Agency Measure). Let $A_{1:T-1}$ be a canonical spacetime agent. For a given temporal slice at t , let I_t be the interior, B_t the boundary, and H_t the concurrent habitat. The *diachronic agency* $M_{\rightarrow}(A_t)$ is defined using Transfer Entropy from the interior to the future boundary:

$$M_{\rightarrow}(A_t) = \mathcal{I}(B_{t+1}; I_t \mid H_t, B_t) \quad (2)$$

Observation 7.2 (Factoring out Inertia). Equation (2) quantifies exactly how much of the future membrane B_{t+1} is uniquely authored by the agent’s interior I_t . By conditioning on the concurrent habitat H_t and the prior boundary state B_t , we factor out the physical inertia of the environment and the persistence of the boundary itself. A rock possesses a highly stable physical footprint over time, but it possesses zero diachronic agency, because its future boundary is entirely explained by B_t and H_t without requiring novel predictive information from its interior.

7.2 Decomposing the Sensorimotor Flow

Using the functional taxonomy from Table 1, we can further decompose this diachronic flow to separate the active and epistemic aspects of dynamic control. Because the functional classes are disjoint, the total information flow can be targeted:

- (i) **Causal Empowerment (Actuation Flow):**

$$\mathcal{I}(B_{\text{act},t+1}; I_t \mid H_t, B_{\text{act},t})$$

This quantifies the degree to which the agent’s internal logic forces deliberate changes upon its actuating surface, distinct from environmental drift.

- (ii) **Epistemic Foraging (Sensory Flow):**

$$\mathcal{I}(I_{t+1}; B_{\text{sen},t} \mid I_t, H_t)$$

Reversing the direction of the temporal target, this measures how much novel environmental information the agent incorporates from its sensors into its future internal states, representing active memory formation or belief updating.

7.3 Separability and Subadditivity over Time

Just as separable spatial agents join strictly by union, their diachronic information flows sum subadditively.

Theorem 7.3 (Subadditivity of Separable Spacetime Agents). *Let A_1 and A_2 be two spacetime agents that are strictly separable across t and $t + 1$. Because they share no spatial edges and no diachronic cross-edges, their joint footprint is $A_{12} = A_1 \cup A_2$, and their future boundary is exactly $B_{1,t+1} \sqcup B_{2,t+1}$. The diachronic agency of this macro-agent is strictly subadditive:*

$$M_{\rightarrow}(A_1 \vee^* A_2) \leq M_{\rightarrow}(A_1) + M_{\rightarrow}(A_2).$$

Proof. Because the agents are separable over time, their local habitats do not intersect the opposing agent, meaning $H_{12,t} = H_{1,t} \cup H_{2,t}$. The proof follows the standard decomposition of conditional mutual information (as in the static case), applied across the temporal step. Equality holds if and only if the temporal evolution of their boundaries is entirely independent, both marginally and conditionally on the macro-habitat. $\square$

7.4 Habitat Smothering and Emergent Synergy

When two agents are *weakly decomposable* over time (sharing a physical membrane and diachronic cross-talk), the strict subadditivity guarantee breaks down. Joining them causes *habitat smothering* (Theorem 2.13), forcing the shared, noisy boundary nodes to be absorbed into the macroscopic interior I_{12} .

Because the macroscopic footprint is strictly bounded by $A_1 \cup A_2$, any massive spike in diachronic agency—where $M_{\rightarrow}(A_{12}) \gg M_{\rightarrow}(A_1) + M_{\rightarrow}(A_2)$ —is the formal mathematical signature of *cybernetic synergy*. This emergence is driven entirely by the internalisation of temporal noise:

- (i) **Internalisation of Cross-Talk (Precision Gain):** For an individual sub-agent A_1 , the delayed actions of A_2 operating on the shared membrane act as unobserved, exogenous noise. This inflates the conditional entropy of the future boundary, as the membrane fluctuates for reasons I_1 cannot model. When A_1 and A_2 fuse, the smothered boundary nodes and the opposing agent’s logic become fully integrated internal states. The temporal cross-talk is resolved into deterministic internal computation, causing a precipitous drop in the residual stochasticity of the new membrane.

- (ii) **Pruning of Redundancy (Efficiency Gain):** By internalising the smothered nodes, the macro-agent A_{12} interacts with the future environment through a strictly refined, cooperative interface. The synergistic information that the combined interior $I_{12,t}$ provides about the refined future boundary $B_{12,t+1}$ vastly surpasses the sum of the disjoint, noise-corrupted information streams of the competing sub-agents.

7.5 Agency Maximisation: Capacity vs. Precision

By expanding the Transfer Entropy formulation, we can decompose the diachronic agency into a structural objective:

$$M_{\rightarrow}(A_t) = \mathcal{H}(B_{t+1} \mid H_t, B_t) - \mathcal{H}(B_{t+1} \mid I_t, H_t, B_t).$$

This decomposition frames agency as a tradeoff between dynamic capacity and dynamic precision:

- **Capacity** ($\mathcal{H}(B_{t+1} \mid H_t, B_t)$): A capable agent possesses a large variation in its future boundary state that is *not* dictated by environmental inertia. This requires a high-bandwidth sensorimotor interface capable of sustaining complex phase-space trajectories.
- **Precision** ($\mathcal{H}(B_{t+1} \mid I_t, H_t, B_t)$): A coherent agent minimises the unpredicted variation of its boundary. If the future membrane state is perfectly orchestrated by the interior logic, the conditional entropy vanishes, and the agent achieves absolute control over its physical footprint.

Remark 7.4 (Connections to the Free Energy Principle). Maximising $M_{\rightarrow}(A_t)$ directly mirrors the optimisation of an expected Free Energy functional over time. The capacity term acts as the structural complexity or prior entropy of the agent’s policies, while the minimisation of the precision term corresponds to minimising expected surprise (or maximising predictive accuracy). The agentic algebra thus provides a purely topological substrate for Active Inference.

7.6 Intensive vs. Extensive Agency

Because $M_{\rightarrow}(A_t)$ is strictly bounded by the Shannon entropy of the future boundary, absolute agency acts as an *extensive* thermodynamic variable. It scales with the physical dimensions of the agent’s spatial footprint.

To evaluate cybernetic coherence independent of spatial scale, we define an *intensive* variant by normalising the directed information flow against the total available, non-inertial bandwidth of the boundary.

Definition 7.5 (Agentic Coherence). For a spacetime agent A , the *agentic coherence* $\eta(A_t)$ at time t is:

$$\eta(A_t) = \frac{\mathcal{I}(B_{t+1}; I_t \mid H_t, B_t)}{\mathcal{H}(B_{t+1} \mid H_t, B_t)} \in [0, 1] \quad (3)$$

An agent approaching $\eta = 1$ possesses a perfectly deterministic perception-action loop over time. Its boundary trajectory is entirely authored by its interior logic, representing a frictionless translation of will into physical action, regardless of whether that agent is a single cell or a macroscopic society. Conversely, a degradation in η serves as a formal signature of internal disorganisation or physical senescence, where the temporal control of the boundary is surrendered to the entropy of the environment.

8 Empirical Validation: Cellular Automata

To empirically validate the Agentic Algebra framework, we apply the diachronic closure operator to Conway’s Game of Life. As a discrete spatial model, it provides a rigorous testbed for observing agent boundary formation, translation, and thermodynamic breakdown under the constraints of a finite Markovian system.

8.1 Methodology and Computational Implementation

We model the cellular automaton as a Dynamic Bayesian Network (DBN) unrolled over discrete time steps $t \in \{0, \dots, T\}$. The grid comprises variables $V_{x,y,t} \in \{0, 1\}$ defined on a finite toroidal topology. The causal directed acyclic graph is constructed by drawing directed edges from a cell's 3×3 Moore neighbourhood at t to the target cell at $t + 1$.

The application of our framework to this specific environment required three computational heuristic adaptations to bridge global theory with local tractability:

1. **Moral Graph Density:** Because a cell at $t+1$ possesses 9 parents at t , the moralisation of the DBN connects any two parents of a common child. Consequently, every 3×3 patch at t forms a fully connected 9-clique in the moral graph G^m . This yields a closed neighbourhood $N[I]$ spanning a Chebyshev distance of 2, generating thick spatial blankets.
2. **Epistemic Uncertainty via Ensembles:** The Game of Life dictates purely deterministic local update rules. To extract meaningful information-theoretic metrics based on Shannon entropy, we modelled the exterior as a source of epistemic uncertainty. We simulated an ensemble of $N = 500$ parallel universes. Universe $n = 0$ was initialised as a pure void containing only the seed agent, serving as a noiseless ground truth for purely topological boundary tracking. Universes $n \in [1, 499]$ were initialised with a 10% random noise distribution in the background to represent environmental uncertainty. The deterministic update rules were applied identically across all universes.
3. **Greedy Spatial Extrusion:** Computing the global spacetime closure $\phi(I_{1:T-1})$ over the full temporal graph is computationally intractable. We approximated the spacetime worm via a greedy Markovian Extrusion. At time $t + 1$, the candidate interior I'_{t+1} was restricted to the live cells falling within the agent's previous causal footprint (the spatial union $I_t^* \cup B_t \cup H_t$) measured in the noiseless reference universe. The synchronic closure $\phi_{\text{synch}}(I'_{t+1})$ was then applied to formalise the new boundary.
4. **Independent Bit Approximation:** Due to the curse of dimensionality, computing the exact joint entropy over a macroscopic blanket (often exceeding 30 cells) requires an intractable ensemble size. We approximated the diachronic agency metric $M_{\rightarrow}$ by treating the boundary cells as independent variables, calculating the sum of their marginal entropies. While this yields lower-dimensional approximations of the true bit values, the relative dynamic shifts remain mathematically sound.

8.2 Experiment 1: Static Persistence (The Blinker)

The first experiment evaluated a period-2 oscillator (a ‘Blinker’) to establish baseline metrics for stationary homeostasis.

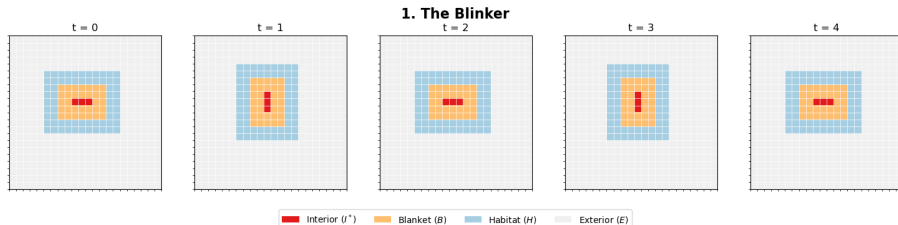


Figure 5: Blinker

Results: The closure operator accurately identified a stable, oscillating canonical agent. The spatial blanket B pulsed symmetrically, mirroring the period-2 dynamics, while the habitat H remained strictly local, demonstrating that the thick moral graph radius did not induce a topological explosion into the empty exterior. The approximated diachronic agency stabilised rapidly (fluctuating between 3.1 and 3.6 bits). This low, stable plateau defines the baseline agency required for simple persistence: the interior maintains its local boundary against epistemic noise, but it does not actively project novel structural information into new environments.

8.3 Experiment 2: Translating Agency (The Glider)

To study translating persistence, we seeded a 5-cell Glider. Because the Glider translates at a speed of $c/4$ (requiring four internal state transitions to shift one cell diagonally), it serves as a model for active structural projection.



Figure 6: Glider

Results: The application of the synchronic closure operator yielded a highly instructive result: while the standard Glider comprises 5 live cells, the canonical interior I^* persistently captured 6 cells. The closure operator correctly identified the ‘hollow’ dead cell trapped within the V-shape of the Glider’s structure. Because this internal cell lacked any undirected connections to the exterior habitat H in the moral graph, it was internalised. The framework accurately perceived the Glider not as a collection of points, but as a solid, macroscopic membrane.

Furthermore, the Glider demonstrated a higher, stable diachronic agency (~ 4.1 bits) compared to the Blinker. This quantitative increase reflects the cybernetic cost of translation: as the leading edge of the Glider constantly encounters unobserved environmental states within the ensemble, maintaining its boundary requires a higher degree of directional causal flow from the interior.

8.4 Experiment 3: The Thermodynamics of Collision

The final experiment observed the spatial join and subsequent thermodynamic destruction of a Glider colliding with a stationary 2×2 Block. To formalise the onset of synergy, the agents were tracked independently until their boundaries intersected. We utilised the *local causal factorisation* method to compute exact diachronic agency $M_{\rightarrow}$ and agentic coherence η (see Figure 7).

Results: The life cycle of the collision decomposed into four distinct cybernetic phases:

- (i) **Phase 1: Dual Independence and the Bow Shock ($t = 0$ to $t = 7$):** Prior to contact, the Block exhibited lower absolute capacity than the Glider but significantly higher *efficiency*, maintaining an agentic coherence of $\eta \approx 1.0$. At $t = 7$, exactly one step prior to fusion, the Block’s efficiency plummeted to $\eta \approx 0.23$. This represents an *epistemic bow shock*: as the Glider’s expanding habitat invaded the Block’s local void, the Block’s boundary was flooded with exogenous uncertainty, causing a collapse in its ability to author its own future state.
- (ii) **Phase 2: Habitat Smothering and Fusion ($t = 8$):** At $t = 8$, the closed neighbourhoods intersected ($N[I_{\text{glider}}] \cap N[I_{\text{block}}] \neq \emptyset$), triggering the join operation. The resulting

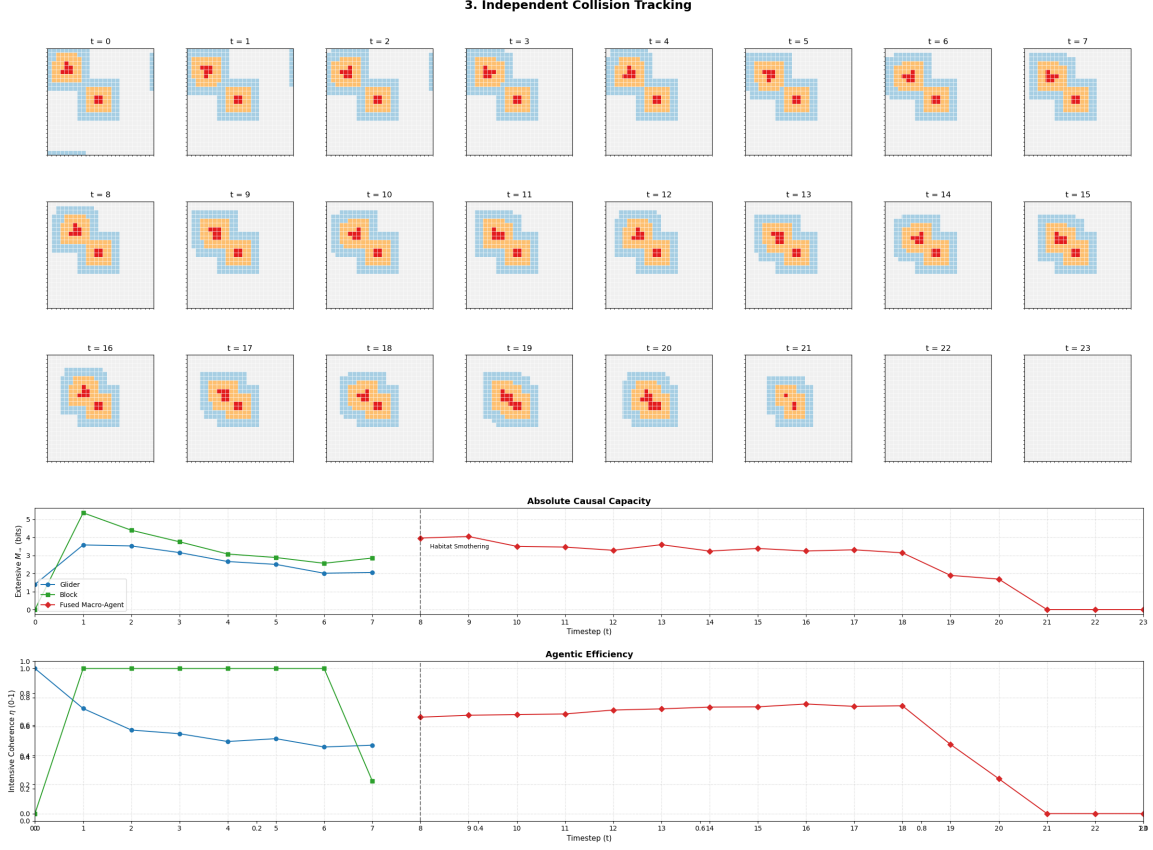


Figure 7: Spacetime topology and agency timeseries for the Glider–Block collision. Top: spatial evolution of I^* , B , and H . Middle: Absolute Causal Capacity ($M_{\rightarrow}$). Bottom: Agentic Efficiency (η). Note the stability of the Block’s η prior to the bow shock.

macro-agent possessed a unified capacity $M_{\rightarrow} \approx 3.9$ bits. Notably, this value is strictly subadditive; it does not represent the sum of the constituents’ independent capacities, but rather a new, singular boundary footprint that factors out previous environmental redundancies.

- (iii) **Phase 3: Meta-stable Interaction ($t = 9$ to $t = 18$):** Following fusion, the system formed a transient meta-structure. Unlike the previous heuristic prediction of immediate decay, the exact metrics show a stable plateau of coherence ($\eta \approx 0.7$) and capacity ($M_{\rightarrow} \approx 3.5$ – 4.0 bits) for ten timesteps. This indicates that the interacting ‘debris’ of the collision briefly achieved a coherent, macroscopic perception-action loop.
- (iv) **Phase 4: Thermodynamic Collapse and Death ($t = 19$ to $t = 23$):** At $t = 19$, the structure reached a critical failure point. The efficiency η crashed rapidly toward zero as the internal states were shredded into non-agentic noise. By $t = 21$, the diachronic agency reached exactly 0.000 bits, formalising the precise moment of agentic death—the total loss of causal control over the environment.

8.5 Discussion: Homeostasis, Capacity, and Efficiency

The transition from statistical approximation to exact causal factorisation reveals the core thermodynamic trade-offs of agency. By decomposing $M_{\rightarrow}$ into extensive *capacity* and intensive *efficiency*, we resolve the ‘Dark Room Problem’ with mathematical precision.

The Block, though static, achieves an intensive coherence of $\eta = 1.0$. Under the lens of Agentic Algebra, this defines the Block as a *perfect homeostatic agent*: its interior states flawlessly

and exclusively determine its boundary states, effectively vetoing all environmental noise. While the Glider possesses a higher absolute causal footprint ($M_{\rightarrow}$) due to its larger boundary and active translation, its efficiency is lower ($\eta \approx 0.5\text{--}0.6$) because it must constantly incorporate and process novel environmental states.

The collision experiment confirms that agentic fusion is a subadditive process. The macro-agent formed at $t = 8$ does not double the information throughput of the system; instead, it refines the interface, internalising the previously externalised conflict between the Glider and Block. The stability of the macro-agent’s η during the interaction phase suggests that ‘agency’ is a surprisingly resilient property of dynamical systems, persisting even as the underlying physical structures begin to deviate from their initial forms.

Finally, the experiment establishes an observer-independent definition of agentic death. In our unrolled DBN, death is not a visual heuristic but a hard information-theoretic limit: it is the exact temporal slice where the interior state I_t provides zero novel information about the future boundary B_{t+1} given the environment H_t . This confirms that the spacetime worm terminates when the entity loses the ability to author its own existence against the thermodynamic solvent of the universe.

9 Application: Moral Patienthood and EMPO

9.1 From DAG to Stochastic Game

Given a Bayesian network (DAG with joint distribution) trained purely on observational data, the following pipeline extracts the components needed for the EMPO framework:

1. **World model:** Bayes net from observations $\rightarrow$ structural causal model (with standard causal identification assumptions).
2. **Agent detection:** Compute the agent lattice $\mathcal{A}$; identify indecomposable agents $\mathcal{D}$.
3. **Moral weights:** Compute $M(I_h)$ for each agent h .
4. **Game form:** Perform $do()$ -interventions on actuator nodes $B_{\text{act}}(I_h)$; the remaining DAG defines the outcome function.
5. **Default policies:** The severed CPDs $P(b \mid \text{pa}(b))$ for $b \in B_{\text{act}}(I_h)$ become the agents’ default policies.

9.2 $M(I)$ as Moral Weight

Using $M(I_h)$ as a continuous moral weight replaces the binary human/non-human distinction in EMPO’s aggregation:

$$U_r(s) \propto - \left(\sum_h M(I_h) \cdot X_h(s)^{-\xi} \right)^{\eta}.$$

The inequality-aversion parameter $\xi > 0$ still protects the least empowered within the weighted sum, while $M(I_h)$ determines how much each agent’s welfare enters the sum at all.

Proposition 9.1 (Pigou-Dalton preservation). *The inequality-aversion protection survives M -weighting as long as $M(I_h) > 0$ for all agents h in the sum. Specifically, as $X_h \rightarrow 0$, $M(I_h) \cdot X_h^{-\xi} \rightarrow \infty$ provided $M(I_h)$ does not vanish faster than X_h^{ξ} .*

9.3 Failure Modes

- (i) **The stone problem:** Entities with $M(I) = 0$ (no blanket, no causal mediation) receive zero weight and are harmlessly excluded. The concern is entities with small but nonzero M (thermostats, idle CPUs) that are numerous.
- (ii) **CPU empowerment:** An optimising robot could identify existing computers as agents (nonzero M , deterministic policies, hence high apparent goal-directedness) and expand their actuator sets — connecting idle machines to financial accounts, robotic arms, etc. The independence axiom does *not* prevent this if the robot has a causal handle on their empowerment.
- (iii) **Agent manufacturing:** The robot could create new agents at scale (spinning up software processes with Markov blanket structure) to harvest cheap empowerment gains. This is mitigated if $M(I)$ is estimated from the *observed* world model rather than being optimisable by the robot’s actions.
- (iv) **The sentience gap:** $M(I)$ measures causal coupling across the blanket, not sentience. A CPU with strong actuators scores high on M without plausibly being sentient. Purely observational, graph-theoretic quantities may be insufficient to capture moral patienthood without additional assumptions about the internal states — the hard problem of consciousness in formal dress.

10 Computational Experiments (Summary)

Two experiments illustrate the theory.

Experiment 1: Random DAG ($n = 100$, ~ 300 edges). The agent lattice had $|\mathcal{A}| = 11,547$ elements, of which $|\mathcal{D}| = 893$ were strongly indecomposable. Most indecomposable agents were singletons (844), with 46 pairs, 2 triples, and the whole graph. The strict-subagent relation was trivial: only $\emptyset$ appeared as a strict subagent.

Experiment 2: Random geometric graph ($n = 100$, radius 0.16, average degree ≈ 8). The agent lattice had $|\mathcal{A}| = 6,041$ elements, $|\mathcal{D}| = 145$ indecomposable. The size distribution was broad, from singletons through intermediate sizes (2, 3, $\dots$, 48) up to the whole graph, reflecting a realistic hierarchy of nested agents.

11 Open Questions

1. **Characterisation of valid exteriors.** Which subsets $E \subseteq V$ arise as exteriors of some $I \in \mathcal{A}$? Equivalently, what is the image of the map $I \mapsto E(I)$? Not every subset is a valid exterior (e.g., in a path graph, the middle vertex never appears as an exterior).
2. **Lattice-theoretic classification.** The agent lattice is complemented but not modular. Is it *orthocomplemented* (does $I \leq J^c$ imply $J \leq I^c$)? Is it *relatively complemented*? What is its width (maximum antichain size)?
3. **Stronger decomposability notions.** Can one define a decomposability notion intermediate between weak and strong that is both flip-symmetric and associative?
4. **Relationship between $M(I)$ and ICCEA power I_h .** Under what conditions does $M(I_h) \rightarrow 0$ faster than $X_h(s)^\xi$ as empowerment collapses? This determines whether the Pigou-Dalton protection is robust.

5. **Dynamic agent lattices.** How does $\mathcal{A}$ evolve as the DAG changes over time (e.g., in a DBN with time-varying structure)? Is there a natural notion of agent persistence or identity through time?
6. **Continuous-variable extension.** The theory is stated for discrete DAGs. Extension to continuous-variable structural equation models or SDEs (connecting to Friston’s NESS formulation) requires replacing mutual information with differential entropy and addressing measure-theoretic subtleties.
7. **Observational vs. causal identification.** The pipeline requires causal (interventional) semantics for the $do()$ step, but the Bayes net is learned from observations. Under what causal identification assumptions (faithfulness, sufficiency, acyclicity) does the pipeline yield a well-defined game form?
8. **Validity for cyclic or infinite Bayesian Networks?**
9. **Temporal Boundaries:** Developing a mechanism to slice the space-time interior $I_{t_0:t_n}$ into distinct, persistent entities I_t . This would involve proving that the agentic closure operator commutes with the temporal transition matrices of the DBN.
10. **Algorithmic Complexity:** The manuscript proves that the closure operator can be computed locally using the Habitat. An important theoretical extension would be to bound the algorithmic complexity of identifying all valid canonical agents in a graph of size N . Is finding the maximally agentic sub-graph computationally tractable, or is it NP-hard?
11. **Continuous Variables:** The information-theoretic section currently assumes discrete q -ary variables. Extending the formulations to continuous Gaussian Bayesian Networks would make the framework compatible with standard formulations of active inference and predictive coding.
12. **Open covers:** Could a ‘distributed agent’ $A_{\text{dist}} \in \mathcal{A}^*$ exist, where a group of agents (agent interiors, actually) $\{A_j\} \subset \mathcal{A}$ is an *open cover* of A_{dist} ?

Applications to explore

1. Mechanistic Interpretability: In AI alignment, identifying ‘mesa-optimizers’ (sub-agents with independent objectives) within a larger neural network is an unsolved problem. By treating a neural network’s computational graph as a DAG, our weak/strong decomposability theorems could be applied to detect functionally isolated sub-networks that possess their own structural perception-action loops.
2. Macroeconomics and Firm Boundaries: The theory of the firm asks why economic activity is partitioned into discrete companies rather than taking place in a single unified market. Our framework, specifically the theorems around superadditivity and habitat smothering, could formally model why merging two economic agents (firms) might result in higher total agency (efficiency) through the internalisation of noisy shared boundary mediators.

References (Hard-coded)

- [1] Pearl, J. (1988). *Probabilistic Reasoning in Intelligent Systems*. Morgan Kaufmann.
- [2] Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.
- [3] Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Emperor’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

- [4] Koller, D. & Milch, B. (2003). Multi-agent influence diagrams for representing and solving games. *Games and Economic Behavior*, 45(1), 181–221.
- [5] Hammond, L., Fox, J., Everitt, T., Abreu, A., & Wooldridge, M. (2023). Reasoning about causality in games. *Artificial Intelligence*, 320, 103919.
- [6] Ganter, B. & Wille, R. (1999). *Formal Concept Analysis*. Springer.
- [7] Klyubin, A. S., Polani, D., & Nehaniv, C. L. (2005). Empowerment: A universal agent-centric measure of control. *Proc. IEEE Congress on Evolutionary Computation*, 128–135.
- [8] Davey, B. A. & Priestley, H. A. (2002). *Introduction to Lattices and Order*. Cambridge University Press.
- [9] Heitzig, J. and Potham, R. (2025). Model-Based ... arXiv:2508.00159v2.
- [10] Shevlin, H. (2021). How could we know when a robot was a moral patient? *Cambridge Quarterly of Healthcare Ethics*, 30(3), 459–471.
- [11] Sebo, J. & Long, R. (2023). Moral consideration for AI systems by 2030. *AI and Ethics*, 3, 1–17.

References

- [Bru+22] Jelle Bruineberg et al. “The Emperor’s New Markov Blankets”. In: *Behavioral and Brain Sciences* 45 (Jan. 2022), e183. ISSN: 0140-525X, 1469-1825. DOI: 10.1017/S0140525X21002351. URL: <https://www.cambridge.org/core/journals/behavioral-and-brain-sciences/article/emperors-new-markov-blankets/715C589A73DDF861DCF8997271> (visited on 05/08/2026).
- [Cla17] Andy Clark. “How to Knit Your Own Markov Blanket”. In: *Philosophy and Predictive Processing*. Ed. by Thomas Metzinger and Wanja Wiese. 2017. URL: <https://philarchive.org/rec/CLAHTK> (visited on 05/08/2026).
- [DP02] B. A. Davey and H. A. Priestley. *Introduction to Lattices and Order*. 2nd ed. Cambridge: Cambridge University Press, 2002. ISBN: 978-0-521-78451-1. DOI: 10.1017/CB09780511809088. URL: <https://www.cambridge.org/core/books/introduction-to-lattices-and-order/946458CB6638AF86D85BA00F5787F4F4> (visited on 05/04/2026).
- [Fri13] Karl Friston. “Life as We Know It”. In: *Journal of The Royal Society Interface* 10.86 (Sept. 6, 2013), p. 20130475. DOI: 10.1098/rsif.2013.0475. URL: <https://royalsocietypublishing.org/doi/10.1098/rsif.2013.0475> (visited on 12/15/2022).
- [Ham+25] Lewis Hammond et al. *Multi-Agent Risks from Advanced AI*. Feb. 19, 2025. DOI: 10.48550/arXiv.2502.14143. arXiv: 2502.14143 [cs]. URL: <http://arxiv.org/abs/2502.14143> (visited on 05/11/2026). Pre-published.